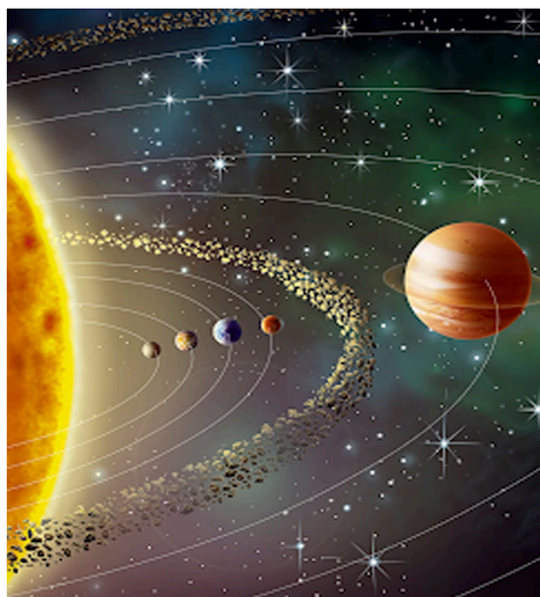


General Knowledge

World Geography

STUDY NOTES

Climatology: Air Masses and Cyclone

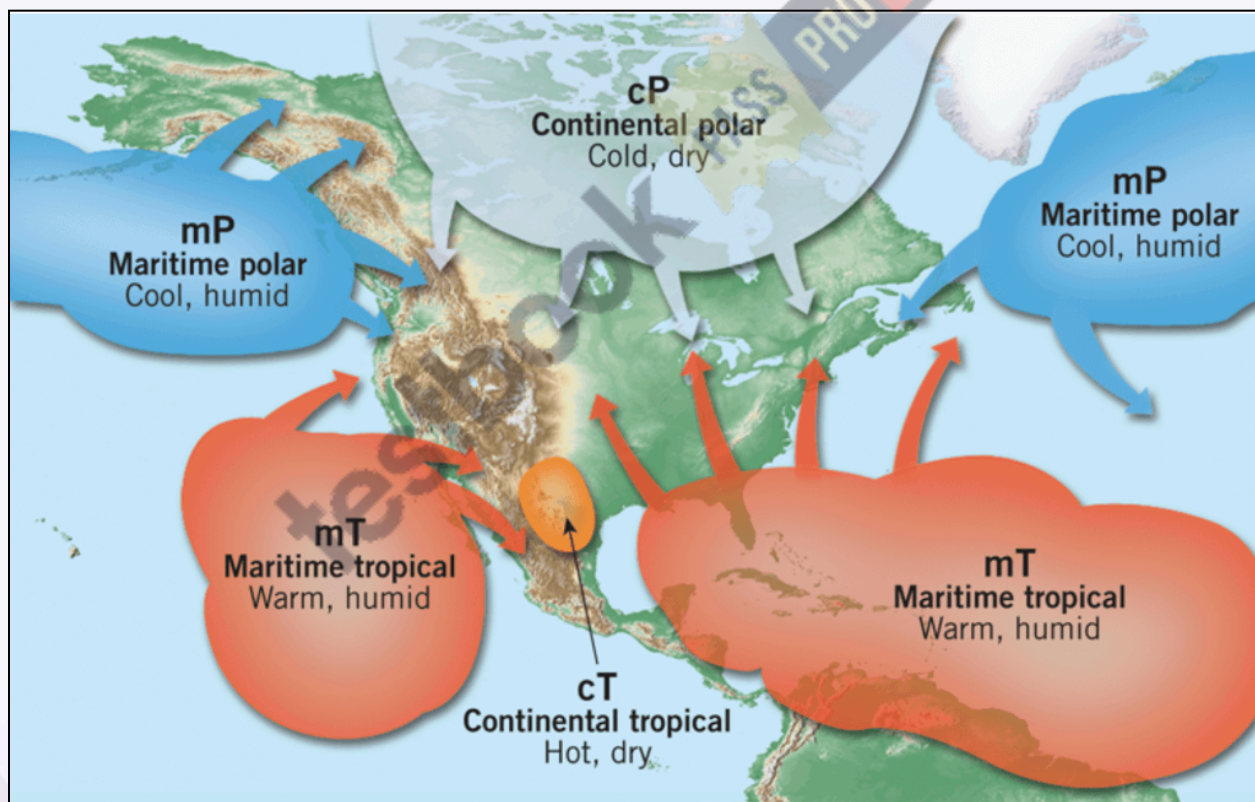


Climatology: Air Masses and Cyclone

Air Mass:

- When the air remains over a homogenous area for a sufficiently longer time, it acquires the characteristics of the area.
- The homogenous regions can be the vast ocean surface or vast plains.
- The air with distinctive characteristics in terms of temperature and humidity is called an **air mass**. It is defined as a large body of air having little horizontal variation in temperature and moisture.
- The homogeneous surfaces, over which the air masses form, are called the **source regions**.

The air masses are classified according to the source regions as:



1. Warm tropical and subtropical oceans
2. The subtropical hot deserts
3. The relatively cold high-latitude oceans

4. The very cold snow-covered continents in high latitudes
5. Permanently ice-covered continents in the Arctic and Antarctica.

Accordingly, the following types of air masses are recognised:

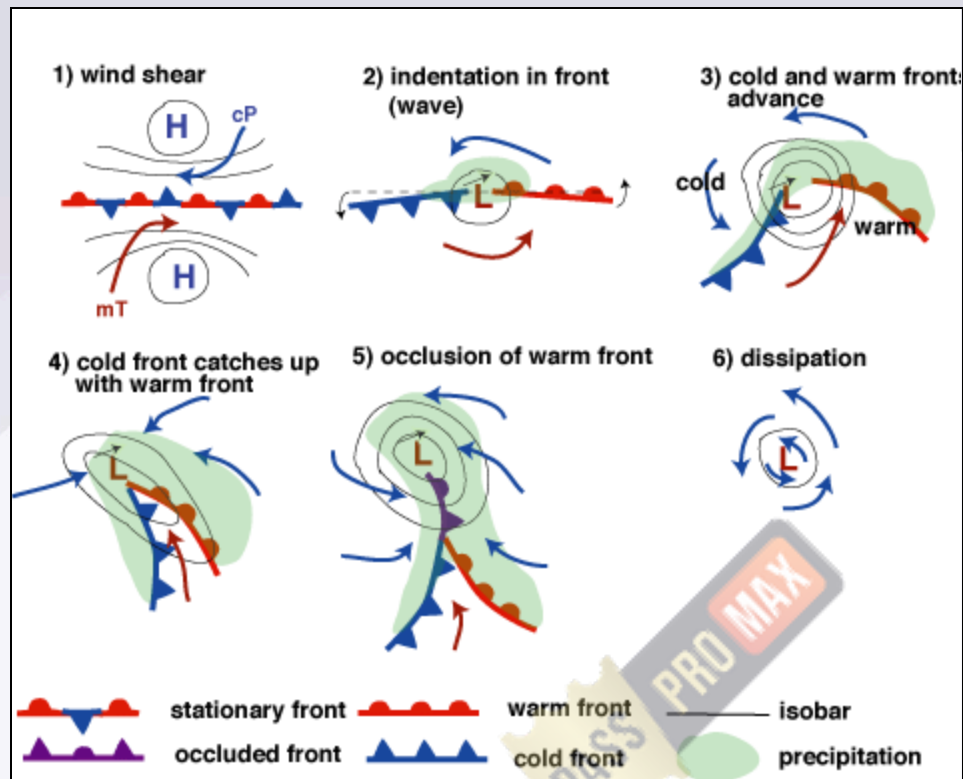
1. Maritime tropical (mT)
2. Continental tropical (cT)
3. Maritime polar (mP)
4. Continental polar (cP)
5. Continental Arctic (cA).

Fronts

- When two different air masses meet, the boundary zone between them is called a front. The process of formation of the fronts is known as **frontogenesis**.
- There are four types of fronts: (a) Cold; (b) Warm; (c) Stationary; and (d) Occluded.
- When the front remains stationary, it is called a **stationary front**.
- When the cold air moves towards the warm air mass, its contact zone is called the **cold front**, whereas if the warm air mass moves towards the cold air mass, the contact zone is a **warm front**.
- If an air mass is fully lifted above the land surface, it is called the **occluded front**.
- Fronts occur in **middle latitudes** and are characterised by steep gradients in temperature and pressure. They bring abrupt changes in temperature and cause the air to rise to form clouds and cause precipitation.

Extra-tropical cyclone:

- The systems developing in the **mid and high latitudes**, beyond the tropics are called the **extratropical cyclones**.
- The **passage of fronts** causes abrupt changes in the weather conditions over the area in the middle and high latitudes.
- **Extratropical cyclones form along the polar front.**



- Initially, the front is stationary. In the northern hemisphere, warm air blows from the south and cold air from the north of the front.
- When the pressure drops along the front, the warm air moves northwards and the cold air moves southward setting in motion an **anticlockwise cyclonic circulation**.
- The cyclonic circulation leads to a well-developed extratropical cyclone, with a warm front and a cold front.
- There are pockets of warm air or warm sector wedged between the forward and the rear cold air or cold sector.
- The warm air glides over the cold air and a sequence of clouds appear over the sky ahead of the warm front and cause precipitation.
- The cold front approaches the warm air from behind and pushes the warm air up. As a result, **cumulus clouds** develop along the cold front.
- The cold front moves faster than the warm front, ultimately overtaking the warm front. The warm air is completely lifted up and the front is occluded and the cyclone dissipates. The processes of wind circulation both at the surface and aloft are closely interlinked.
- The extratropical cyclone differs from the tropical cyclone in a number of ways. The extra tropical cyclones have a **clear frontal system** which is not present in the tropical cyclones.

- They **cover a larger area** and can **originate over the land and sea**. Whereas tropical cyclones originate only over the seas and on reaching the land they dissipate.
- The wind velocity in a tropical cyclone is much higher and it is more destructive.
- The extra tropical cyclones **move from west to east** but tropical cyclones move from east to west.

Tropical Cyclones:

- Tropical cyclones are **violent storms** that originate over oceans in tropical areas and move over to the **coastal areas** bringing about large-scale destruction caused by **violent winds, very heavy rainfall and storm surges**.
- They are known as **Cyclones in the Indian Ocean, Hurricanes in the Atlantic, Typhoons in the Western Pacific and South China Sea, and Willy-willies in Western Australia**.

Region	Name of Cyclone
Indian Ocean	Cyclones
Atlantic Ocean	Hurricanes
Western Pacific & South China Sea	Typhoons
Western Australia	Willy-willies

- Tropical cyclones originate and intensify over warm tropical oceans. The conditions favourable for the formation and intensification of tropical storms are:

(i) Large sea surface with a temperature higher than 27° C

(ii) Presence of the Coriolis force

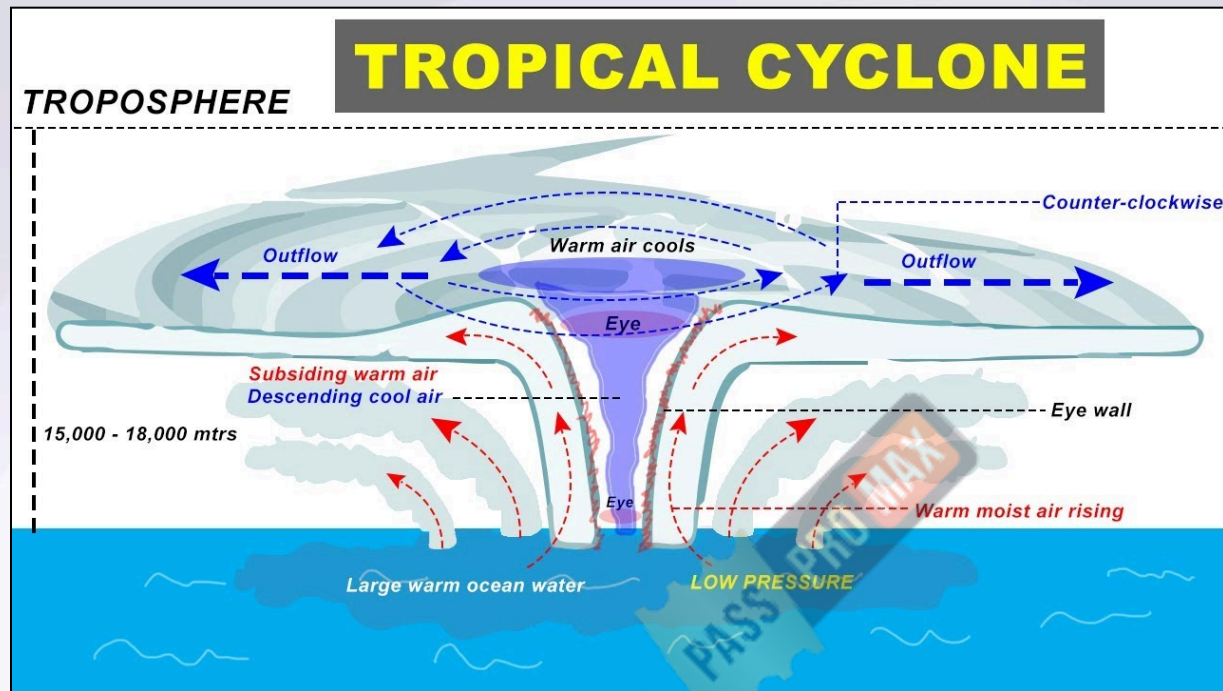
(iii) Small variations in the vertical wind speed

(iv) A pre-existing weak low-pressure area or low-level cyclonic circulation

(v) Upper divergence above the sea level system.

- The energy that intensifies the storm, comes from the **condensation process in the cumulonimbus clouds**, surrounding the centre of the storm. With a **continuous supply of moisture from the sea**, the storm is further strengthened. On reaching the land the moisture supply is cut off and the storm dissipates. The place where a tropical cyclone crosses the

coast is called the **landfall of the cyclone**. The **cyclones which cross 20°N latitude generally, recurve and are more destructive.**



- A mature tropical cyclone is characterised by the **strong spirally circulating wind around the centre, called the eye.**
- The diameter of the circulating system can vary between 150 and 250 km. The eye is a region of calm with subsiding air.
- Around the eye is the **eye wall**, where there is a strong spiralling ascent of air to a greater height reaching the tropopause.
- The wind reaches maximum velocity in this region, reaching as high as 250 km per hour. Torrential rain occurs here.
- From the eye wall rain bands may radiate and trains of cumulus and cumulonimbus clouds may drift into the outer region.
- The diameter of the storm over the **Bay of Bengal, Arabian Sea and Indian Ocean** is between **600 - 1200 km.**
- The system moves slowly about **300 - 500 km per day.** The cyclone creates storm surges and they inundate the coastal lowlands. The storm peters out on the land.

Naming of cyclones:

The cyclones that are formed in any ocean basin around the world are named by the **Regional Specialised Meteorological Centres (RSMCs) and Tropical Cyclone Warning Centres (TCWCs)**.

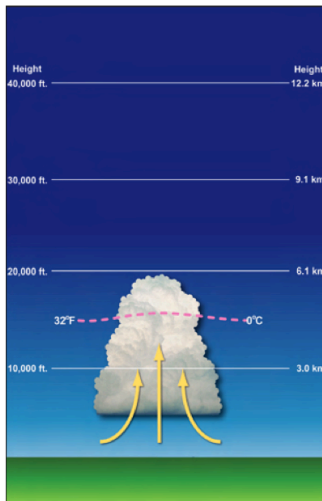
- There are a total of **six RSMCs in the world**, including the India Meteorological Department (IMD).
- **The World Meteorological Organization (WMO) and the United Nations Economic and Social Commission for the Asia Pacific (ESCAP) have been naming cyclonic storms since 2000.**
- **The India Meteorological Department (IMD)** names the cyclones developing over the north Indian Ocean, including the Bay of Bengal and the Arabian Sea.
- **It also issues advisories to 12 other nations** in the region on the development of cyclones and storms.
- In 2000, a group of nations called WMO/ESCAP-- **Bangladesh, India, the Maldives, Myanmar, Oman, Pakistan, Sri Lanka and Thailand** - decided to name cyclones in the region.
- In 2018, five more countries were added - **Iran, Qatar, Saudi Arabia, United Arab Emirates and Yemen.**
- After the aforementioned countries sent in suggestions, the WMO/ESCAP Panel on Tropical Cyclones (PTC) finalized the list.

Thunderstorms & Tornadoes

- These local winds are of short duration, occurring over a small area but are violent.
- Thunderstorms are caused by **intense convection on moist hot days**. A thunderstorm is a well-grown **cumulonimbus cloud** producing thunder and lightning.
- When the clouds extend to heights where sub-zero temperature prevails, hails are formed and they come down as **hailstorms**. If there is insufficient moisture, a thunderstorm can generate **dust storms**.
- A thunderstorm is characterised by an intense updraft of rising warm air, which causes the clouds to grow bigger and rise to a greater height.
- This causes precipitation. Later, downdraft brings down to earth the cool air and the rain.

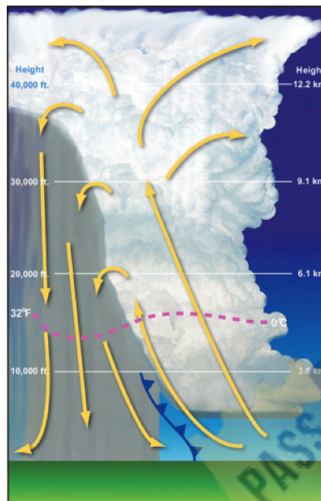
- From severe thunderstorms sometimes spiralling wind descends with very low pressure at the centre, causing massive destruction on its way. Such a phenomenon is called a **tornado**.
- Tornadoes generally occur in middle latitudes and if they occur over the sea, they are called **waterspouts**.

The Thunderstorm Life Cycle



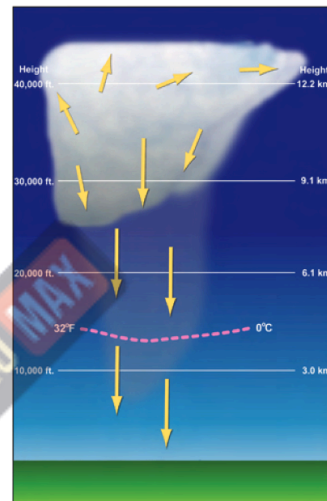
Developing Stage

- Towering cumulus cloud indicates rising air
- Usually little if any rain during this stage
- Lasts about 10 minutes
- Occasional lightning



Mature Stage

- Most likely time for hail, heavy rain, frequent lightning, strong winds, and tornadoes
- Storm occasionally has a black or dark green appearance
- Lasts an average of 10 to 20 minutes but some storms may last much longer



Dissipating Stage

- Downdrafts, downward flowing air, dominate the storm
- Rainfall decreases in intensity
- Can still produce a burst of strong winds
- Lightning remains a danger

